

```
(setq scilab-commands '(
  "endfunction" "continue" "function" "apropos"
  "elseif" "resume" "return" "select"
  "abort" "break" "catch" "clear"
  "pause" "while" "case" "else"
  "exit" "help" "quit" "then"
  "what" "clc" "end" "for"
  "pwd" "try" "who" "do"
  "if"
))
```

Each keyword category is then submitted to GNU Emacs in the following order: scilab-xcosfuns, scilab-functions, scilab-primtives, scilab-commands, scilab-variables. Some modifications to the syntax table are also necessary to specify the comments highlighting and that some characters can be a keyword or a part of a keyword:

```
(modify-syntax-entry ?\/ ". 12b" synTable)
(modify-syntax-entry ?\n "> b" synTable)
(modify-syntax-entry ?_ "w" synTable)
(modify-syntax-entry ?! "w" synTable)
```

As a result, I have the Scilab code highlighted with the default colours: five colours for the keywords, one colour for the strings and one colour for the comments. The default colours are quite different from each other, so further improvements are not necessary. Now all the keywords are recognised without any error. Last, add (require 'scilab-mode) to your dotemacs file. This syntax highlighting file for GNU Emacs has been made following the notes written by Xah Lee (see Reference 5). The full *scilab-mode.el* code is available on GitHub.

Vim

In the case of Vim, the keywords in each file are sorted only alphabetically, which is not necessary. I can also leave the keywords in the order in which they were originally extracted. Then merge every four lines (to reduce the number of rows) and add some Vim script code. Even in this case, a short example is given by the commands:

```
syn keyword sciCommands abort apropos break case
syn keyword sciCommands catch clc clear continue
syn keyword sciCommands quit resume return select
syn keyword sciCommands endfunction exit for function
syn keyword sciCommands help if pause pwd
syn keyword sciCommands quit resume return select
syn keyword sciCommands then try what while
syn keyword sciCommands who
```

Now it's necessary to specify that some characters can be a keyword or a part of a keyword:

```
setlocal iskeyword+=|.-!
```

```
setlocal iskeyword+=$-$
setlocal iskeyword+=%%
```

Last, some Vim script code about the colours is added. As a result, I have the Scilab code highlighted with seven custom colours:

```
hi Operator      guifg=#0000CD " Medium blue
hi Conditional   guifg=#DC143C " Crimson
hi Statement     guifg=#FF8C00 " Dark orange
hi Function      guifg=#1E90FF " Dodger blue
hi Label         guifg=#D2B48C " Tan
hi String        guifg=#808080 " Gray
hi Comment       guifg=#3CB371 " Medium sea green
```

```
Hilink sciPrimitives Operator
Hilink sciCommands Conditional
Hilink sciVariables Statement
Hilink sciFunctions Function
Hilink sciXcosfuns Label
```

My syntax highlighting file for Vim is partially based on those written by Václav Mocek (see Reference 6) and Patricio Toledo (Reference 7). The full *scilab.vim* code is available on GitHub.

Scilab has some interesting LaTeX capabilities about plots and data management. A syntax highlighting file is fairly easy to build for GNU Emacs and Vim. It's easy also for JEdit (Java based and cross-platform) and Notepad++ (Windows only). The interaction between Scilab and GNU Emacs, in a manner similar to ESS for R, is possible but does not really interest me, so I haven't explored it yet. This article is the last of a series of four articles about Scilab, but I do think that, "Scilab is a rare gem that is waiting to be discovered more broadly", as stated by Raphaël Auphan, the CEO of Scilab (see Reference 8). 

References

- [1] <http://forge.scilab.org/index.php/p/latexmath/>, last visited on 25/04/2015.
- [2] <http://math.tex.sourceforge.net>, last visited on 25/04/2015.
- [3] http://en.wikipedia.org/wiki/Witch_of_Agnesi, last visited on 25/04/2014.
- [4] Fusco, Bianchetti, Rosnati, *Chimica organica* (Organic chemistry), vol. 1, Guadagni, Milan, 1980.
- [5] http://ergoemacs.org/emacs/elsp_syntax_coloring.html, last visited on 25/04/2014.
- [6] http://www.vim.org/scripts/script.php?script_id=1137, last visited on 25/04/2014.
- [7] http://www.vim.org/scripts/script.php?script_id=396, last visited on 25/04/2014.
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